Program Summary - Personal Project 3 SAS Code.sas

Execution Environment

Author:	u64322043
File:	/home/u64322043/Personal Project 3 - Average Price Based on the Factors in India/Personal Project 3 SAS Code.sas
SAS Platform:	Linux LIN X64 5.14.0-284.30.1.el9_2.x86_64
SAS Host:	ODAWS01-USW2-2.ODA.SAS.COM
SAS Version:	9.04.01M8P02222023
SAS Locale:	en_US
Submission Time:	3/17/2026, 11:17:58 PM
Browser Host:	99.209.173.182
User Agent:	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/145.0.0.0 Safari/537.36
Application Server:	ODAMID00-USW2-2.ODA.SAS.COM

Code: Personal Project 3 SAS Code.sas

```
/* Sicheng Lu */
* Business Analytics Post Grad;
* Bachelor of Economics Honours with Minor in Statistics;
* Personal Project 3: Flight Prices based on the factors in India;
* SAS Regression Modelling;

%web_drop_table(WORK.IMPORT);
/* Importing the data file to this SAS */

FILENAME REFFILE '/home/u64322043/Personal Project 3 - Average Price Based on the Factors in India/Clean_Dataset.csv';

PROC IMPORT DATAFILE=REFFILE
    DBMS=CSV
    OUT=flight_prices_india;
    GETNAMES=YES;
RUN;
* Double check the data has been uploaded successfully;
PROC CONTENTS DATA=flight_prices_india; RUN;
/* Printing the table for the first 10 datasets */
proc print data=flight_prices_india (obs=10);
title 'The first 10 datasets';
* In order to do modelling regression,
all category variable will convert to numerical variable;
data updated_data;
set flight_prices_india;
if airline = 'Vistara' then airline_num = 1;
if airline = 'Air India' then airline_num = 2;
if airline = 'SpiceJet' then airline_num = 3;
if airline = 'GO_FIRST' then airline_num = 4;
if airline = 'Indigo' then airline_num = 5;
if airline = 'AirAsia' then airline_num = 6;

if stops = 'zero' then num_stops = 0;
if stops = 'one' then num_stops = 1;
if stops = 'two' then num_stops = 2;

if destination_city = 'Bangalore' then dest = 1;
if destination_city = 'Chennai' then dest = 2;
if destination_city = 'Delhi' then dest = 3;
if destination_city = 'Hyderabad' then dest = 4;
if destination_city = 'Kolkata' then dest = 5;
if destination_city = 'Mumbai' then dest = 6;

if class = 'Busines' then classes = 1;
if class = 'Economy' then classes = 2;

run;
* Regression Modelling;
proc reg data = updated_data;
```

```
    model price = airline_num num_stops dest classes days_left;
    title 'Regression Model for Average Price Based on the Factors';
run;

* SAS Graph;
proc sgplot data=updated_data;
   reg y=price x=days_left / group = classes;
   title "Flight Price vs Days Left by Class";
run;
* class = 1 = Busines = Red
* class = 2 = Economy = Blue
```

Log: Personal Project 3 SAS Code.sas

Warnings (1)

Notes (15)

```
1          OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
68
69         /* Sicheng Lu */
70         * Business Analytics Post Grad;
71         * Bachelor of Economics Honours with Minor in Statistics;
72         * Personal Project 3: Flight Prices based on the factors in India;
73         * SAS Regression Modelling;
74
75         %web_drop_table(WORK.IMPORT);
76         /* Importing the data file to this SAS */
77
78         FILENAME REFFILE '/home/u64322043/Personal Project 3 - Average Price Based on the Factors in India/Clean_Dataset.csv';
79
80         PROC IMPORT DATAFILE=REFFILE
81         DBMS=CSV
82         OUT=flight_prices_india;
83         GETNAMES=YES;
84         RUN;

NOTE: Import cancelled.  Output dataset WORK.FLIGHT_PRICES_INDIA already exists.  Specify REPLACE option to overwrite it.
NOTE: The SAS System stopped processing this step because of errors.
NOTE: PROCEDURE IMPORT used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              305.40k
      OS Memory           29348.00k
      Timestamp           03/18/2026 03:17:57 AM
      Step Count                        370  Switch Count  0
      Page Faults                       0
      Page Reclaims                     14
      Page Swaps                        0
      Voluntary Context Switches        0
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           0

85         * Double check the data has been uploaded successfully;


86         PROC CONTENTS DATA=flight_prices_india; RUN;

NOTE: PROCEDURE CONTENTS used (Total process time):
      real time           0.02 seconds
      user cpu time       0.03 seconds
      system cpu time     0.00 seconds
      memory              3365.50k
      OS Memory           30636.00k
      Timestamp           03/18/2026 03:17:57 AM
      Step Count                        371  Switch Count  0
      Page Faults                       0
      Page Reclaims                     294
      Page Swaps                        0
      Voluntary Context Switches        3
      Involuntary Context Switches      3
```

```
      Block Input Operations            0
      Block Output Operations           16


           /* Printing the table for the first 10 datasets */
           proc print data=flight_prices_india (obs=10);
           title 'The first 10 datasets';
           * In order to do modelling regression,
           all category variable will convert to numerical variable;

NOTE: There were 10 observations read from the data set WORK.FLIGHT_PRICES_INDIA.
NOTE: PROCEDURE PRINT used (Total process time):
      real time           0.02 seconds
      user cpu time       0.02 seconds
      system cpu time     0.00 seconds
      memory              2394.50k
      OS Memory           30376.00k
      Timestamp           03/18/2026 03:17:57 AM
      Step Count                        372  Switch Count  1
      Page Faults                       0
      Page Reclaims                     253
      Page Swaps                        0
      Voluntary Context Switches        10
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           16


           data updated_data;
           set flight_prices_india;
           if airline = 'Vistara' then airline_num = 1;
           if airline = 'Air India' then airline_num = 2;
           if airline = 'SpiceJet' then airline_num = 3;
           if airline = 'GO_FIRST' then airline_num = 4;
           if airline = 'Indigo' then airline_num = 5;
           if airline = 'AirAsia' then airline_num = 6;

           if stops = 'zero' then num_stops = 0;
           if stops = 'one' then num_stops = 1;
           if stops = 'two' then num_stops = 2;

           if destination_city = 'Bangalore' then dest = 1;
           if destination_city = 'Chennai' then dest = 2;
           if destination_city = 'Delhi' then dest = 3;
           if destination_city = 'Hyderabad' then dest = 4;
           if destination_city = 'Kolkata' then dest = 5;
           if destination_city = 'Mumbai' then dest = 6;

           if class = 'Busines' then classes = 1;
           if class = 'Economy' then classes = 2;

           run;

NOTE: There were 300153 observations read from the data set WORK.FLIGHT_PRICES_INDIA.
NOTE: The data set WORK.UPDATED_DATA has 300153 observations and 16 variables.
NOTE: DATA statement used (Total process time):
      real time           0.06 seconds
      user cpu time       0.04 seconds
      system cpu time     0.02 seconds
      memory              3576.09k
      OS Memory           32428.00k
      Timestamp           03/18/2026 03:17:57 AM
      Step Count                        373  Switch Count  6
      Page Faults                       0
      Page Reclaims                     528
      Page Swaps                        0
      Voluntary Context Switches        27
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           75272


           * Regression Modelling;
           proc reg data = updated_data;
           model price = airline_num num_stops dest classes days_left;
           title 'Regression Model for Average Price Based on the Factors';
           run;
```

```
WARNING: ODS graphics with more than 5000 points have been suppressed. Use the PLOTS(MAXPOINTS= ) option in the PROC REG statement
         to change or override the cutoff.
121
122        * SAS Graph;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.10 seconds
      user cpu time       0.07 seconds
      system cpu time     0.04 seconds
      memory              5157.56k
      OS Memory           33984.00k
      Timestamp           03/18/2026 03:17:57 AM
      Step Count                        374  Switch Count  7
      Page Faults                       0
      Page Reclaims                     742
      Page Swaps                        0
      Voluntary Context Switches        59
      Involuntary Context Switches      3
      Block Input Operations            0
      Block Output Operations           110144


123        proc sgplot data=updated_data;
124           reg y=price x=days_left / group = classes;
125           title "Flight Price vs Days Left by Class";
126        run;

NOTE: PROCEDURE SGPLOT used (Total process time):
      real time           1.20 seconds
      user cpu time       0.78 seconds
      system cpu time     0.07 seconds
      memory              23401.40k
      OS Memory           50072.00k
      Timestamp           03/18/2026 03:17:58 AM
      Step Count                        375  Switch Count  41
      Page Faults                       0
      Page Reclaims                     10210
      Page Swaps                        0
      Voluntary Context Switches        347
      Involuntary Context Switches      11
      Block Input Operations            0
      Block Output Operations           86152

NOTE: Marker and line antialiasing has been disabled for at least one plot because the threshold has been reached. You can set
      ANTIALIASMAX=300200 in the ODS GRAPHICS statement to enable antialiasing for all plots.
NOTE: There were 300153 observations read from the data set WORK.UPDATED_DATA.

127        * class = 1 = Busines = Red
128        * class = 2 = Economy = Blue
129
130
131
132        OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
133        ODS HTML CLOSE;
134        &GRAPHTERM; ;*';*";*/;RUN;QUIT;
135        QUIT;RUN;
136        ODS HTML5 (ID=WEB) CLOSE;
137
138        FILENAME _GSFNAME;
NOTE: Fileref _GSFNAME has been deassigned.
139        DATA _NULL_;
140        RUN;

NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              471.34k
      OS Memory           29100.00k
      Timestamp           03/18/2026 03:17:58 AM
      Step Count                        376  Switch Count  0
      Page Faults                       0
      Page Reclaims                     23
      Page Swaps                        0
      Voluntary Context Switches        0
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           0
```

```
OPTIONS NOTES STIMER SOURCE SYNTAXCHECK;
```

Results: Personal Project 3 SAS Code.sas

The CONTENTS Procedure

Data Set Name	WORK.FLIGHT_PRICES_INDIA	**Observations**	300153
Member Type	DATA	**Variables**	12
Engine	V9	**Indexes**	0
Created	03/17/2026 22:34:23	**Observation Length**	96
Last Modified	03/17/2026 22:34:23	**Deleted Observations**	0
Protection		**Compressed**	NO
Data Set Type		**Sorted**	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	221
First Data Page	1
Max Obs per Page	1363
Obs in First Data Page	1327
Number of Data Set Repairs	0
Filename	/saswork/SAS_work8AEC0000C4AB_odaws02-usw2-2.oda.sas.com/SAS_work26A40000C4AB_odaws02-usw2-2.oda.sas.com/flight_prices_india.sas7bdat
Release Created	9.0401M8
Host Created	Linux
Inode Number	1946180355
Access Permission	rw-r--r--
Owner Name	u64322043
File Size	28MB
File Size (bytes)	29097984

Alphabetic List of Variables and Attributes

#	Variable	Type	Len	Format	Informat
1	VAR1	Num	8	BEST12.	BEST32.
2	airline	Char	9	$9.	$9.
7	arrival_time	Char	13	$13.	$13.
9	class	Char	7	$7.	$7.
11	days_left	Num	8	BEST12.	BEST32.
5	departure_time	Char	13	$13.	$13.
8	destination_city	Char	6	$6.	$6.
10	duration	Num	8	BEST12.	BEST32.
3	flight	Char	7	$7.	$7.
12	price	Num	8	BEST12.	BEST32.
4	source_city	Char	5	$5.	$5.
6	stops	Char	4	$4.	$4.

The first 10 datasets

Obs	VAR1	airline	flight	source_city	departure_time	stops	arrival_time	destination_city	class	duration	days_left	price
1	0	SpiceJet	SG-8709	Delhi	Evening	zero	Night	Mumbai	Economy	2.17	1	5953
2	1	SpiceJet	SG-8157	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	2.33	1	5953
3	2	AirAsia	I5-764	Delhi	Early_Morning	zero	Early_Morning	Mumbai	Economy	2.17	1	5956
4	3	Vistara	UK-995	Delhi	Morning	zero	Afternoon	Mumbai	Economy	2.25	1	5955
5	4	Vistara	UK-963	Delhi	Morning	zero	Morning	Mumbai	Economy	2.33	1	5955
6	5	Vistara	UK-945	Delhi	Morning	zero	Afternoon	Mumbai	Economy	2.33	1	5955
7	6	Vistara	UK-927	Delhi	Morning	zero	Morning	Mumbai	Economy	2.08	1	6060
8	7	Vistara	UK-951	Delhi	Afternoon	zero	Evening	Mumbai	Economy	2.17	1	6060
9	8	GO_FIRST	G8-334	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	2.17	1	5954
10	9	GO_FIRST	G8-336	Delhi	Afternoon	zero	Evening	Mumbai	Economy	2.25	1	5954

Regression Model for Average Price Based on the Factors

The REG Procedure
Model: MODEL1
Dependent Variable: price

Number of Observations Read	300153
Number of Observations Used	82405
Number of Observations with Missing Values	217748

Analysis of Variance					
Source	**DF**	**Sum of Squares**	**Mean Square**	**F Value**	**Pr > F**
Model	5	3.874667E13	7.749335E12	150006	<.0001
Error	82399	4.256752E12	51660241		
Corrected Total	82404	4.300343E13			

Root MSE	7187.50588	**R-Square**	0.9010
Dependent Mean	19022	**Adj R-Sq**	0.9010
Coeff Var	37.78500		

Parameter Estimates					
Variable	**DF**	**Parameter Estimate**	**Standard Error**	**t Value**	**Pr > |t|**
Intercept	1	96493	148.11895	651.46	<.0001
airline_num	1	-355.49930	15.39379	-23.09	<.0001
num_stops	1	8192.39549	65.79045	124.52	<.0001
dest	1	404.70180	16.74447	24.17	<.0001
classes	1	-47000	65.71229	-715.23	<.0001
days_left	1	-138.64953	1.85673	-74.67	<.0001